

# 304 SS RIS and Oxidation Prediction by Off-Lattice KMC Model

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## 1 Introduction

## 2 Simulation Method

In this study, an off-lattice kinetic Monte Carlo (olKMC) model is employed to simulate radiation-induced segregation (RIS) and grain boundary oxidation in 304 stainless steel (304 SS). The simulation framework consists of two main components: stress- and irradiation-modulated elemental diffusion, and grain boundary oxidation driven by oxygen adsorption and in-plane diffusion. In the following subsections, the model geometry and species are first introduced, followed by the KMC algorithm, the RIS driving force, and the oxidation kinetics.

### 2.1 Model Overview

The simulation domain is a three-dimensional cubic box of edge length  $L = 1$  mm, representing the grain interior of a polycrystalline 304 stainless steel specimen. A planar grain boundary (GB) is embedded in the domain, defined by the equation

$$Ax + By + Cz + D = 0 \quad (1)$$

where the coefficients  $(A, B, C, D)$  control the orientation and position of the GB. Periodic boundary conditions are applied in the direction parallel to the GB plane for oxygen diffusion, while metal atoms are reflected at the box boundaries.

Four particle species are tracked throughout the simulation, as summarised in Table 1.

Table 1: Particle species in the simulation.

| Species | Type   | Role                                                            |
|---------|--------|-----------------------------------------------------------------|
| Si      | solute | diffuses in grain interior; segregates to GB under irradiation  |
| Cr      | solute | diffuses in grain interior; segregates to GB under irradiation  |
| Ni      | solute | diffuses in grain interior; segregates to GB under irradiation  |
| O       | oxygen | adsorbs at the GB surface line and diffuses within the GB plane |

Initial positions of Si, Cr, and Ni atoms are loaded from a pre-generated configuration file that represents a realistic grain microstructure. A static stress field  $\sigma(\mathbf{r})$ , obtained from external finite-element calculations, is interpolated to each atomic position at runtime. All simulations are performed at  $T = 673$  K (400 °C), representative of light water reactor operating conditions.

## 2.2 KMC Algorithm and Hop Rate

The simulation follows the Bortz–Kalos–Lebowitz (BKL) variant of the kinetic Monte Carlo algorithm. At each step, a list of all possible events is maintained together with their rates (propensities). The total rate is

$$R = \sum_j r_j \quad (2)$$

where the sum runs over all events  $j$  in the current event pool. The physical time increment associated with each KMC step is drawn from an exponential distribution:

$$\Delta t = -\frac{\ln u_1}{R}, \quad u_1 \sim \mathcal{U}(0, 1) \quad (3)$$

An event  $k$  is then selected with probability  $r_k/R$  by drawing a second uniform random number  $u_2$  and finding the index  $k$  such that

$$\sum_{j < k} r_j < u_2 R \leq \sum_{j \leq k} r_j \quad (4)$$

After executing event  $k$ , all propensities are recalculated and the procedure repeats.

**Event types.** The event pool contains five types of events: random-walk diffusion of Si, Cr, and Ni atoms (`SILI_DIFF`, `CROM_DIFF`, `NIK_DIFF`); oxygen adsorption at the GB surface line (`OXYGEN_ABS`); and in-plane oxygen diffusion within the GB (`OXYGEN_DIFF`). Stochastic oxidation is evaluated after each diffusion event, as described in Section ??.

**Metal-atom hop rate.** Each metal-atom diffusion event is governed by a stress- and irradiation-modified Arrhenius rate:

$$r_i = \nu_0 \exp\left(-\frac{\Delta E + \text{tr}(\boldsymbol{\sigma}) \Omega - \Delta\mu_i(x)}{k_B T}\right) \times s(d_{\text{GB}})^{-2} \quad (5)$$

where  $\nu_0 = 10^{13} \text{ s}^{-1}$  is the attempt frequency,  $\Delta E = 1.0 \text{ eV}$  is the migration barrier,  $\text{tr}(\boldsymbol{\sigma})$  is the local hydrostatic stress trace interpolated from the external stress field,  $\Omega = 10^{-28} \text{ m}^3$  is the activation volume, and  $\Delta\mu_i(x)$  is the irradiation-induced effective chemical potential described in Section 2.3. The factor  $s(d_{\text{GB}})^{-2}$  accounts for spatially adaptive resolution, as described below.

**Adaptive spatial resolution.** To resolve the steep concentration gradients near the GB while keeping the bulk computationally tractable, the jump distance and hop rate are jointly scaled by a position-dependent scalar  $s(d_{\text{GB}})$ , defined via a sigmoid function of the distance  $d_{\text{GB}}$  to the grain boundary:

$$s(d_{\text{GB}}) = 10^{\frac{-1}{1 + \exp\left(\frac{d_{\text{GB}} - d_0}{\ell}\right)}} \quad (6)$$

where  $d_0 = 30 \mu\text{m}$  is the crossover distance and  $\ell$  is the length scale of the transition. The scalar ranges from  $s = 0.1$  at the GB to  $s = 1$  far away. The actual jump distance and propensity used at each step are

$$\delta_{\text{actual}} = \delta_0 s, \quad r_{\text{actual}} = r_0 s^{-2} \quad (7)$$

where  $\delta_0$  and  $r_0$  are the base jump distance and rate, respectively. Because diffusivity  $D \propto r \delta^2$ , the product  $r_{\text{actual}} \delta_{\text{actual}}^2 = r_0 \delta_0^2$  is preserved, ensuring that the physical diffusivity is unchanged across the domain while the GB region is sampled at up to  $10\times$  finer spatial resolution.

## 2.3 Radiation Induced Segregation

To phenomenologically capture RIS, we employ an effective chemical potential that accounts for the preferential enrichment or depletion of alloying elements to grain boundaries under irradiation. The diffusion flux of alloying species driven by the chemical potential gradient is expressed as

$$J_i = -\frac{D_i C_i}{k_B T} \frac{\partial (\mu_0 + \Delta\mu_i(x))}{\partial x} \quad (8)$$

where  $J_i$  is the diffusion flux of species  $i$ ,  $D_i$  is the diffusion coefficient,  $C_i$  is the concentration,  $k_B$  is the Boltzmann constant,  $T$  is the absolute temperature,  $\mu_0$  is the reference chemical potential, and  $\Delta\mu_i(x)$  is the radiation-induced effective chemical potential contribution.

The fluxes satisfy the mass-balance constraint

$$\sum_i J_i + J_V = 0 \quad (9)$$

where  $J_V$  is the vacancy flux, given by Fick's first law:

$$J_V = -D_V \frac{\partial C_V}{\partial x} \quad (10)$$

with  $D_V$  and  $C_V$  denoting the vacancy diffusion coefficient and vacancy concentration, respectively.

For irradiated 304 stainless steel (304 SS), vacancies are continuously generated in the grain interior, whereas the grain boundary serves as a strong vacancy sink. Accordingly, the vacancy concentration profile near the grain boundary is described by

$$C_V(x) = C_V^\infty(\text{dose}) - (C_V^\infty(\text{dose}) - C_V^{\text{GB}}) \exp(-kx) \quad (11)$$

where  $x$  is the distance from the grain boundary,  $C_V^\infty(\text{dose})$  is the bulk vacancy concentration at a given irradiation dose,  $C_V^{\text{GB}}$  is the vacancy concentration at the grain boundary, and  $k$  is the decay constant. In the present work,  $C_V^{\text{GB}} = 0$  is assumed. Thus the vacancy flux is within the form

$$J_V = -k D_V C_V^\infty(\text{dose}) \exp(-kx) \quad (12)$$

The mass-balance constraint (Eq. 9) can be rewritten as:

$$\sum_i J_i = k D_V C_V^\infty(\text{dose}) \exp(-kx) \quad (13)$$

By intergrating the two sides of 13, the mass-balance constraint for RIS effective chemical potential can be expressed as:

$$\sum_i D_i C_i \Delta\mu_i(x) = K_B T D_V C_V^\infty (\text{dose}) \exp(-kx) + \text{Const} \quad (14)$$

We assume that the effective chemical potential is zero at infinite distance from the grain boundary, thus the constant term in Eq. 14 is zero. The effective chemical potential constraint by mass balance can be expressed as :

$$\sum_i D_i C_i \Delta\mu_i(x) = K_B T D_V C_V^\infty (\text{dose}) \exp(-kx) \quad (15)$$

The effective chemical potential contribution for each alloying element is assumed to follow an exponential decay form:

$$\Delta\mu_i(x) = \Delta\mu_i^{\text{GB}} \exp\left(-\frac{x}{\lambda}\right) \quad (16)$$

where  $\Delta\mu_i^{\text{GB}}$  is the effective chemical potential contribution at the grain boundary, and  $\lambda$  is the characteristic length scale. For 304 SS, the segregation of Ni and Cr are mainly contributed by Inverse Kirkendall effect,  $\Delta\mu_i^{\text{GB}}$  is given by:

$$\Delta\mu_i^{\text{GB}} = \Delta\mu_0^{\text{GB}} \ln \frac{D_i}{D_{Fe}} \quad (17)$$

where  $D_{Fe}$  is the diffusion coefficient of Fe, and  $\Delta\mu_0^{\text{GB}}$  is a fitting parameter that represents the overall magnitude of the effective chemical potential at the grain boundary.  $D_{Cr} > D_{Fe} > D_{Ni}$ , therefore  $\Delta\mu_{Cr}^{\text{GB}} > 0$  and  $\Delta\mu_{Ni}^{\text{GB}} < 0$ , which captures the depletion of Cr and enrichment of Ni at the grain boundary under irradiation. For Si, the segregation behavior is treated separately, because in addition to the vacancy-mediated inverse Kirkendall effect, Si can also migrate through the formation of solute-interstitial complexes. To account for this additional mechanism, the effective chemical potential contribution of Si is decomposed into a vacancy-related part and an interstitial-related part:

$$\Delta\mu_{Si}(x) = \Delta\mu_{Si}^V(x) + \Delta\mu_{Si}^I(x) \quad (18)$$

where  $\Delta\mu_{Si}^V(x)$  denotes the vacancy-mediated contribution and  $\Delta\mu_{Si}^I(x)$  denotes the contribution associated with solute-interstitial complexes.

The vacancy-mediated contribution is assumed to follow the same functional form as that of substitutional alloying elements:

$$\Delta\mu_{Si}^V(x) = \Delta\mu_{Si}^{V,GB} \exp\left(-\frac{x}{\lambda_V}\right) \quad (19)$$

where  $\Delta\mu_{Si}^{V,GB}$  is the vacancy-mediated effective chemical potential of Si at the grain boundary and  $\lambda_V$  is the corresponding characteristic length scale.

The contribution from solute-interstitial complexes is expressed as

$$\Delta\mu_{Si}^I(x) = \Delta\mu_{Si}^{I,GB} \exp\left(-\frac{x}{\lambda_I}\right) \quad (20)$$

where  $\Delta\mu_{Si}^{I,GB}$  is the effective chemical potential contribution at the grain boundary associated with interstitial-complex transport, and  $\lambda_I$  is the characteristic decay length of this contribution.

Accordingly, the total effective chemical potential for Si becomes

$$\Delta\mu_{Si}(x) = \Delta\mu_{Si}^{V,GB} \exp\left(-\frac{x}{\lambda_V}\right) + \Delta\mu_{Si}^{I,GB} \exp\left(-\frac{x}{\lambda_I}\right) \quad (21)$$

In the present model, the interstitial-complex contribution is assumed to dominate the RIS behavior of Si, i.e.,

$$\left|\Delta\mu_{Si}^{I,GB}\right| > \left|\Delta\mu_{Si}^{V,GB}\right| \quad (22)$$

so that the net driving force reproduces the experimentally observed segregation tendency of Si at grain boundaries under irradiation. For simplicity, if  $\lambda_V$  and  $\lambda_I$  are taken to be identical, Eq. 21 can be rewritten as

$$\Delta\mu_{Si}(x) = \left(\Delta\mu_{Si}^{V,GB} + \Delta\mu_{Si}^{I,GB}\right) \exp\left(-\frac{x}{\lambda}\right) \quad (23)$$

where the combined grain-boundary effective chemical potential term is fitted to yield the correct Si segregation behavior.